

Modeling and sensorless control of an electromagnetic valve actuator

Peter Eyabi ^{a,*}, Gregory Washington ^b

^a Eaton Aerospace, Grand Rapids, MI 49512, United States

^b The Ohio State University, Columbus, OH 43210, United States

Accepted 22 November 2005

Abstract

This paper presents the modeling and control of an electromagnetic valve actuator. The model is nonlinear and takes into account secondary nonlinearities like hysteresis, saturation, bounce and mutual inductance. Experimental methods for identifying parameters of the model are presented. Sliding mode control is utilized to reduce the seating velocity of the plunger as it strikes the core housing. This reduces actuator noise due to the high impact velocity of the plunger. In order to eliminate the need for position and velocity sensors, a nonlinear observer is developed that only employs coil current measurement. The position estimate is used as feedback to track a desired trajectory. The control objective is to minimize energy consumption and to reduce the seating velocity, which should improve actuator fatigue life and reduce impact noise. Simulation and experimental results are presented for an actuator designed and built in our laboratory. The experimental results show very good agreement with the analytical model.

© 2006 Elsevier Ltd. All rights reserved.

Keywords: Electro magnetic actuator; Sliding mode; Sensorless; Variable valve actuation; Nonlinear control

1. Introduction

Electromagnetic actuators (EMA) are increasingly becoming the actuator of choice in industry, due to their ruggedness, low cost, reduced complexity, relative high force density and ease of control. Solenoids in particular are being used in fuel injection actuation, exhaust gas recirculation (EGR) systems, food dispensers, refrigerators, washing machines, etc.

One of the latest applications of solenoid based EMA's include electromagnetic valve (EMV) actuation systems (see Fig. 1). The EMV configuration that will be examined in this study consists of two solenoids with a common plunger and opposing springs (Fig. 1). These devices are presently being developed for implementation of advanced combustion strategies in internal combustion engines. The main issue with deployment of EMV's in internal combustion engines stems from the noise produced when the energized plunger strikes the core. Other challenges

include, but are not limited to, control of the seating velocity, improved energy consumption, trajectory shaping with a minimum number of measurements and high actuation speeds. It is therefore necessary to have tighter control tolerances and more in-depth models of this latest generation EMA. Additionally, it is imperative that the overall system be cost effective. This means that the system may have to optimize performance with fewer available sensors. These demands, coupled with the strongly nonlinear dynamics of the EMA, make the use of classical sensor-based control strategies a less attractive option.

The EMV is one of the promising solutions to the challenge of reduction in fuel consumption and vehicle emissions. The idea of “individual cylinder control” and camless engines has reinvigorated interest in the concept of variable valve timing (VVT) or fully flexible valve actuation systems (FFVA). This effectively means that there is direct control of both valve timing and valve lift. This methodology provides another controlled degree of freedom to both the intake and exhaust dynamics and has the added potential of significantly improving engine torque and overall fuel economy.

* Corresponding author.

E-mail address: peyabi@yahoo.com (P. Eyabi).

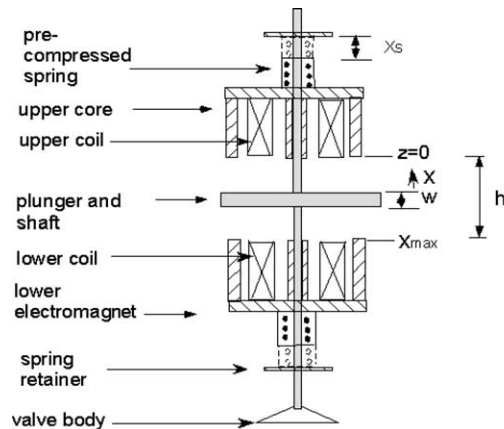


Fig. 1. Schematic of the EMV and actual device.

The plunger-striking problem, which may contribute to reduced structural integrity as well as noise, can be addressed by reducing the plunger seating velocity (plunger speed just before it impacts the core or housing of each electromagnet). Plunger seating velocity reduction can be obtained partly by mechanical design and entirely by electronic control. In addition, a nonlinear EMV model will be developed and validated. This model will include secondary nonlinearities like saturation, hysteresis, bounce and mutual induction. These nonlinearities are important in modeling the electromagnetic force to a reasonable degree of accuracy since the force exhibits these characteristics.

The control of an EMV entails modulating some measured mechanical variable like the velocity or position. Certain mechanical variables may not be measured accurately due to operational or environmental conditions. From a perspective of cost it is also advantageous to use the least number of sensors possible. Hence, there is a need to utilize other signals, usually an electrical variable, from which information on certain mechanical variables could be inferred. This is termed *sensorless estimation*. A primary objective in this paper is the modeling and sensorless control of seating velocity for a class of EMV actuators. The controls solution is a nonlinear model-based approach that is invariant with respect to disturbances and/or model uncertainties [9].

To date, a significant amount of work has been conducted on EMV's [1–15]. A few notable studies related to the present work are presented here. Lequesne [10] presented a configuration of the EMV, where the magnetic force is produced by two permanent magnets and a pair of coils. Modeling of eddy currents was taken into account since this represents core losses. Their effect was modeled by finite element. The finite element model was validated with experimental data although some mismatches existed. Wang and Stefanopoulou [3] arrived at a model, which takes into account the linear and saturation dynamics of the coil. Simulation data was then validated with experimental data obtained from an engine equipped with an EMV. Sensor resolution reduced the accuracy of the numerical approximation of the velocity. An attempt was

made to solve the soft seating problem of the EMV actuator with a linear controller and observer of an unknown structure. The control algorithm was based on feeding back the position measurement to a gain-scheduled tracking controller. Control voltages were as high as 100 V. Hoffman and Stefanopoulou [6] used the validated model in [3] and designed a tracking controller for valve position that comprised a feedforward and feedback controller. The position and velocity were estimated from position measurements. The feedback controller design was based on the state feedback method and this used the position measurements for control. Tai [2] proposed a model that was based on a linear magnetic model. The model included a valve lash. The H_∞ methodology was used to design a tracking controller based on position. Seating velocity was reduced to an acceptable value.

Butzman et al. [4] attempted a nonobserver based sensorless control algorithm based on current measurement. The sensorless control strategy was based on observing the ratio of the current through the coil and its derivative, which is then compared to a predefined value. The coil current is increased or decreased appropriately, according to the error. The algorithm was later modified to adapt the energy input for each event based on the landing performance of the armature in the previous event. The algorithm was tested on a combustion engine, and landing velocities of 0.2–0.3 m/s were reported. However no comparative data was presented on the effectiveness of the estimator to accurately estimate the position and velocity signal.

The work presented in this study seeks to include a comprehensive model incorporating nonlinear effects such as bounce, mutual inductance and hysteresis. Thus the research is a logical progression of the previously mentioned work. The model is then used to optimize the design an EMV that reduces energy consumption to the point where it is 42 V ready. A set of design rules are then developed to aid in design of low energy systems. In addition, an observer-based sensor-less control strategy is used to address the seating velocity problem. Finally, the results are experimentally validated with an EMV optimized to reduce power consumption.

2. Modeling of the EMV

An EMV was designed and built as shown in Fig. 1. It comprises two opposing electromagnets with pre-compressed coil springs at both ends, and a plunger midway between the electromagnets. The moving mass is composed of the plunger, plunger shaft, spring and spring retainer as shown in Fig. 1.

2.1. Mechanical model

The derivation of the mechanical model is well documented in [14]. The free body diagram in Fig. 2 was used in the derivation of the mechanical dynamics, where F_{mu} and F_{md} are the forces due to the upper and lower electromagnet respectively, F_f is the force due to dry friction, F_{su} and F_{sd} are the spring forces of the upper and lower springs, respectively, and F_{bu} and F_{bd} are the damping forces of the upper and lower springs, respectively.

The differential equation describing the motion of the plunger is

$$M_t \ddot{z} = -2K_s(z - y) - B\dot{z} - C \operatorname{sgn}(\dot{z}) - F_h - F_{md} + F_{mu} \quad (1)$$

where K_s is the spring constant, F_{md} and F_{mu} are the magnetic force of the lower and upper electromagnets, respectively, B is the lumped damping coefficient of the upper and lower spring, and z is the displacement of the lumped mass, M_t

$$z = y - X \quad \text{and} \quad y = \frac{h - w}{2} = \text{const.} \quad (2a)$$

$$0 \leq z \leq z_{\max} \quad \text{and} \quad -X_{\max} \leq X \leq X_{\max}$$

Also, the air-gaps are defined as

$$x_u = y - X \quad \text{and} \quad x_d = y + X \quad (2b)$$

The force due to the hard nonlinearity (core face) is modeled as

$$F_h = \begin{cases} K_u(\max(z, z_{\max})) + B_u \dot{z} & z \geq z_{\max} \\ K_d(\min(z, z_{\min})) + B_d \dot{z} & z \leq z_{\min} \end{cases} \quad (3)$$

where F_h is the force due to the hard nonlinearity, K is the stiffness of the electromagnet core, B is its damping coefficient, w is the width of the plunger, x_s is the length of spring pre-compression, x is the air-gap (x_u for upper electromagnet and x_d for lower electromagnet) and the subscripts u and d represent the upper and lower electromagnet, respectively. Disturbances due to engine forces are neglected in this derivation.

2.2. Electrical model

The differential equation describing the electrical subsystem dynamics is given in (4)

$$\frac{d\lambda}{dt} = -f_3(\lambda, x, V_L)R + V_s \quad (4)$$

where λ is the flux linkage, V_s is the supply voltage, R is the coil resistance and V_L is the voltage across the inductor.

A schematic describing the current relationship is shown in Fig. 3 and the current function is defined in (5).

$$f_3(\lambda, x, V_L) = i = i_r + i_d = f(x, \lambda) + f_1(V_L) - \tau_j \frac{di_d}{dt}. \quad (5)$$

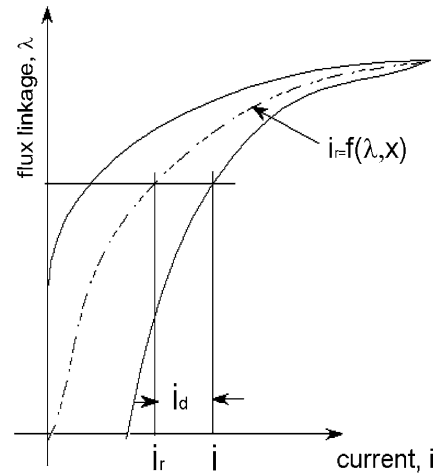


Fig. 3. Magnetization curve showing relationship of current functions.

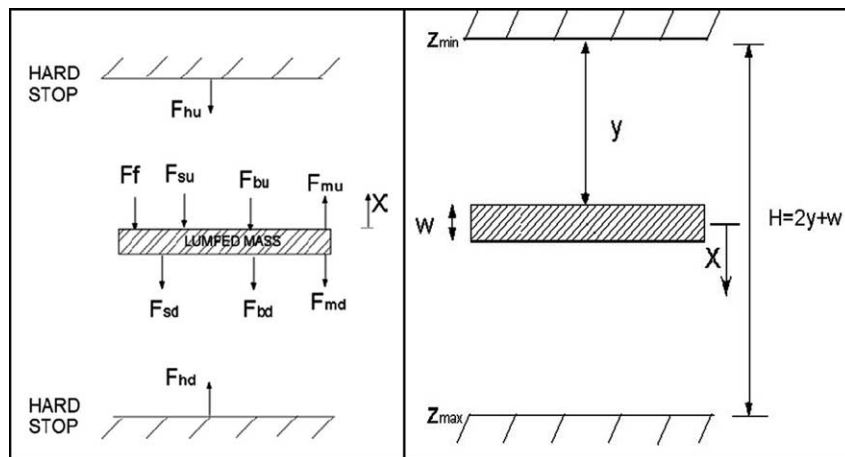


Fig. 2. Free-body diagram (right) and motion space (left).

4. Experimental validation

4.1. Magnetic subsystem

4.1.1. Flux linkage

The flux linkage is calculated using Eq. (4). The magnetization curve for the electromagnet was plotted for different air-gaps and different voltages, and validated against experimental data, as shown in Figs. 5 and 6.

The experimental and simulated plots show a close match at different voltages and air-gaps, showing that saturation and hysteresis has been adequately modeled for various operating points of the electromagnet (Figs. 5 and 6).

4.1.2. Force

The magnetic force was modeled as shown in Eq. (11).

$$F_m = m_2 \lambda^2 + m_1 \lambda + m_0 \quad (11)$$

The parameters m_2 and m_1 are dependent on air-gap, so they were plotted against air-gap and the best fit was obtained using a polynomial of the form

$$m_i = p_{i8}x^8 + p_{i7}x^7 + p_{i6}x^6 + p_{i5}x^5 + p_{i4}x^4 + p_{i3}x^3 + p_{i2}x^2 + p_{i1}x + p_{i0} \quad (12)$$

The regression coefficients were then used in a simulation and the theoretical force was validated with experimental data as shown in Fig. 7.

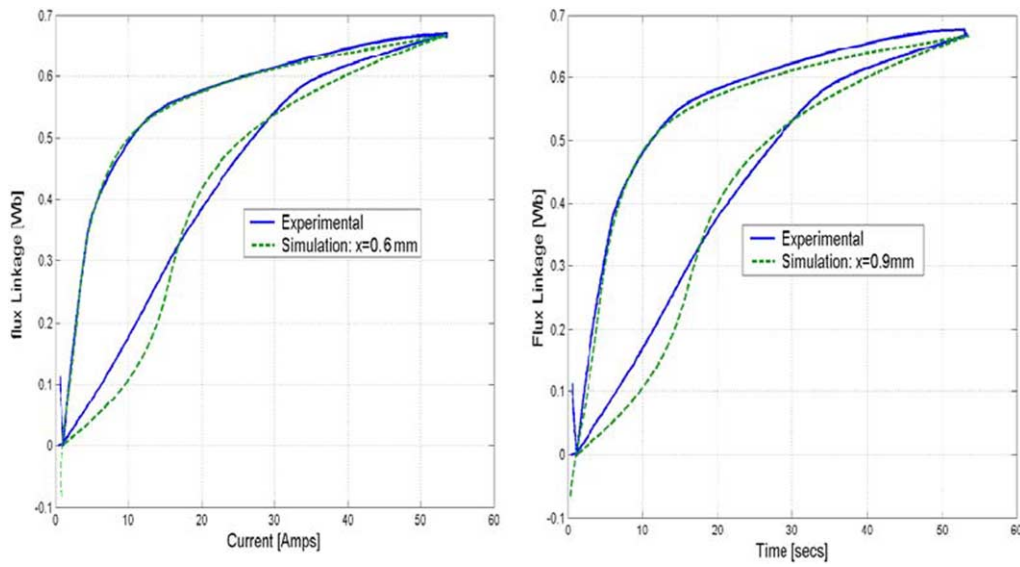


Fig. 5. Magnetization curve for different air-gaps at 35.5 V.

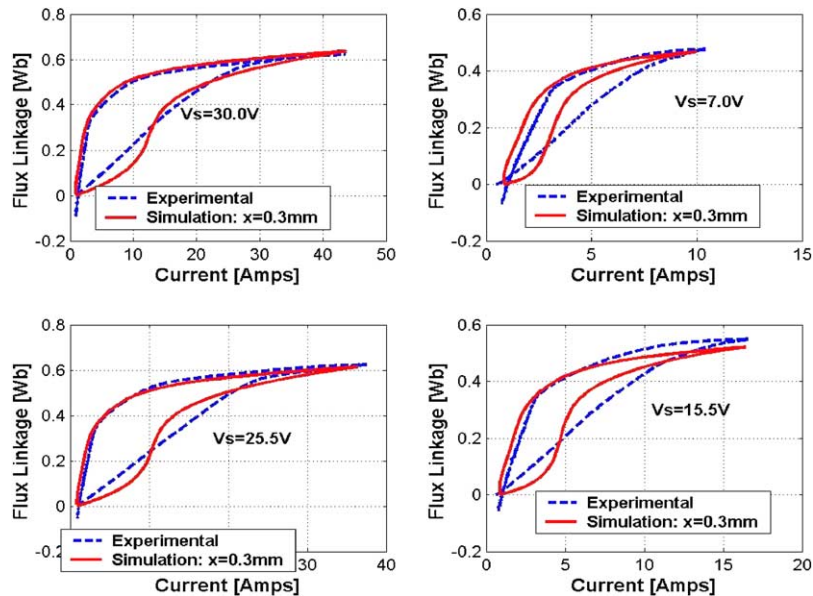


Fig. 6. Magnetization curve for different voltages at 0.3 mm air-gap.

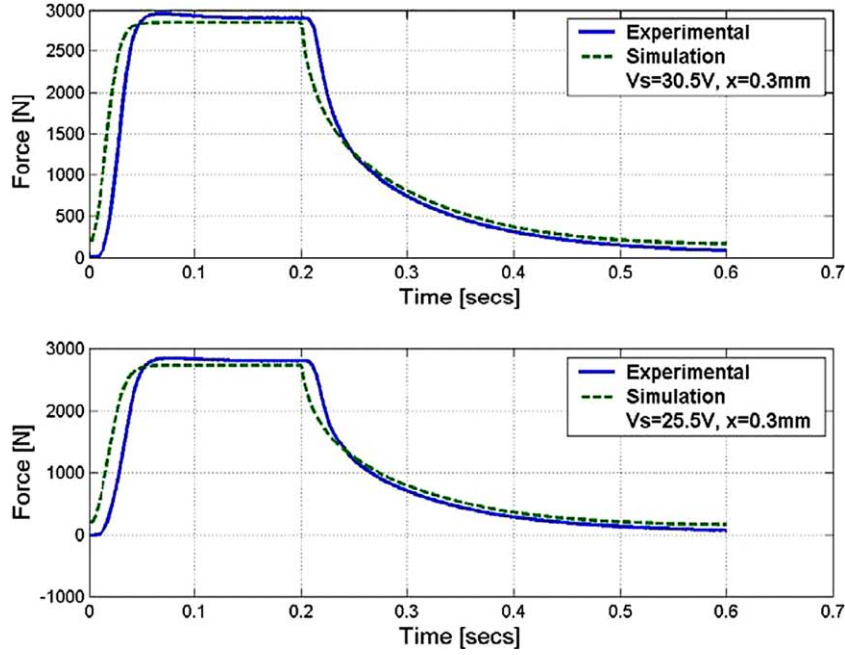


Fig. 7. Magnetic force at different voltages (air-gap constant).

4.2. Electrical subsystem

4.2.1. Current

The *restoring current* for each air-gap was obtained by fitting a mean magnetization curve to the already generated hysteresis curve, in the least square sense. A good fit was obtained using a polynomial of the form

$$i_r = g_4 \lambda^4 + g_3 \lambda^3 + g_2 \lambda^2 + g_1 \lambda + g_0 \quad (13)$$

where, g_4 , g_3 , g_2 and g_1 are coefficients dependent on the air-gap. The effect of the coefficients of (13) can be represented by plotting them against the air-gap and fitting a curve to them in the least square sense. The best fit was given by (14)

$$g_i = n_{i6}x^6 + n_{i5}x^5 + n_{i4}x^4 + n_{i3}x^3 + n_{i2}x^2 + n_{i1}x + n_{i0} \quad (14)$$

The restoring function could be obtained by using Eq. (13). The dissipation function (6) for each air-gap was estimated by first estimating the dissipation function $f_1(V_L)$, and the time constant τ . Using the restoring function already identified, i_d was calculated from Eq. (5) using experimental data. Values of the parameters d_{11} , d_{12} , d_{21} , d_{22} and τ were incorporated into Eq. (7) and the simulation results were compared to the experimental results of i_d for different air-gaps. This was done iteratively using a direct search optimization technique until a good fit was obtained. Hence, the parameters of Eq. (7) were then identified. These parameter values were then used in a simulation. The plot of the numerical and experimental data for voltage and current for different air-gaps and voltages is shown in Figs. 8 and 9. There is a close agreement between experimental data and model for a wide range of operating points. The slight discrepancy for small voltages arises

from the fact that it was difficult to capture the transient excursions of the voltage across the inductor. The polynomial order selected was based on a polynomial that would give the best fit for different data samples representing different operational conditions of the actuator.

4.3. Mechanical subsystem

From Eq. (1) the parameters to be identified from the mechanical dynamics include M , B , and C .

These parameters were identified using methods outlined in [16]. The parameters obtained were used in a simulated model of the mechanical subsystem with an initial condition input. The results were plotted along with the experimental data of the three system (Fig. 10).

4.4. EMV system

After validating each subsystem, the entire device was assembled and both coils were energized with an open loop voltage. The data was plotted against that obtained from a simulation of the model as shown in Figs. 11a and 11b. There is a good fit between experimental data and simulation, except during bounce, as seen in the plots of coil current and velocity.

Also, the fidelity of the model was dependent on the accurate estimation of the air-gap at the equilibrium position of the plunger, with none of the electromagnets energized. Hence this directly affected the accuracy of the estimated force and the state estimates. The equilibrium position is never at zero, due to residual flux on the electromagnets, but the model assumes that the plunger equilibrium position is zero. The equilibrium position was

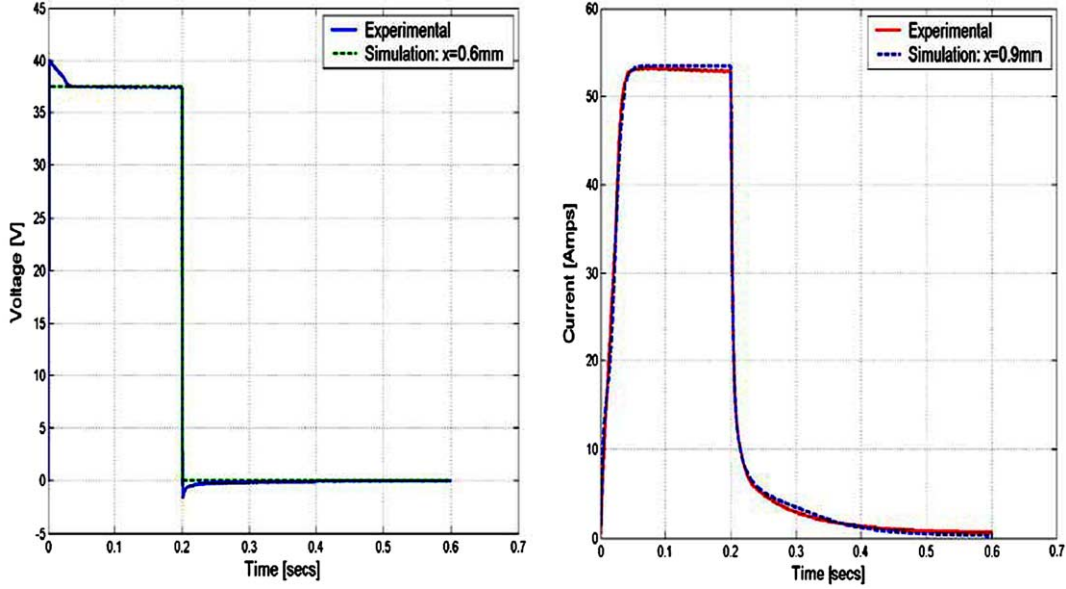


Fig. 8. Validation: voltage (left), current (right).

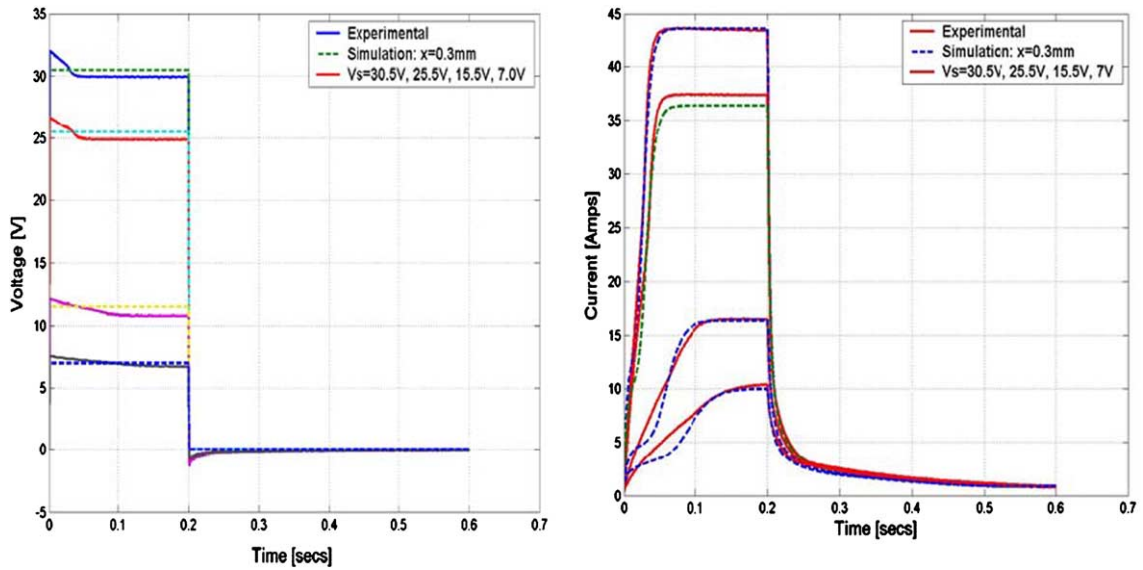


Fig. 9. Validation for different voltages at 0.3 mm air-gap: voltage (left), current (right).

observed to be anywhere from -0.3 mm to $+0.3\text{ mm}$. This value is crucial since an error of 0.1 mm might cause the estimated force to differ by as much as 200 N at small air-gaps and this will influence the validation errors of force, flux and velocity.

5. Sensorless observer design for EMV

It is assumed that only the current measurement is available and it is intended to estimate the position and velocity of the plunger. Therefore an observer will be designed that will drive the estimation error of the position and the velocity to zero in finite time, at which point the state estimates will be equal to their actual values.

The sliding mode methodology was used here, due to its robustness to uncertainty and ease of implementation [9,14]. The observer structure for the system in (10) is as follows:

$$\begin{aligned}
 \dot{\hat{z}}_1 &= \hat{z}_2 - M_1 \text{sign}(S_1) - M_2 \text{sign}(S_2) \\
 \dot{\hat{z}}_2 &= \frac{1}{M_t} (-2K_s(\hat{z}_1 - y) - B\hat{z}_2 - C \text{sgn}(\hat{z}_2) - \hat{F}_h - \hat{F}_{md} + \hat{F}_{mu}) \\
 &\quad - M_3 \text{sign}(S_1) - M_4 \text{sign}(S_2) \\
 \dot{\hat{\lambda}}_u &= -f_{3u}(\hat{\lambda}_u, \hat{x}_u, V_{Lu})R + V_{su} - M_5 \text{sign}(s_1) \\
 \dot{\hat{\lambda}}_d &= -f_{3d}(\hat{\lambda}_d, \hat{x}_d, V_{Ld})R + V_{sd} - M_6 \text{sign}(s_2)
 \end{aligned} \tag{15}$$

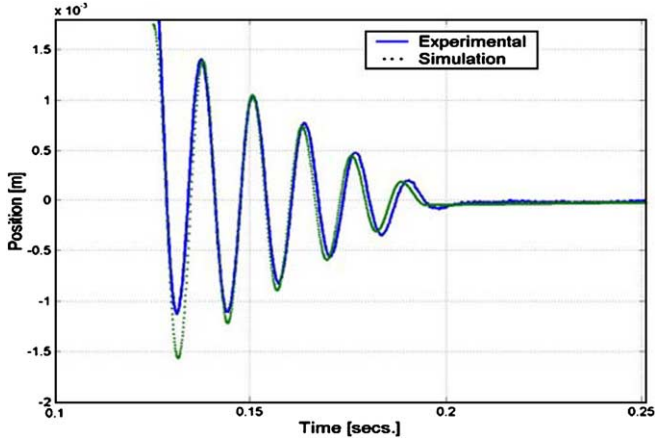


Fig. 10. Experimental validation of free motion.

where, the sliding manifolds are defined as

$$s_1 = \hat{i}_u - i_u \quad \text{and} \quad s_2 = \hat{i}_d - i_d \quad (16)$$

For sufficiently large gains, M_i ($i = 1, \dots, 6$), sliding mode is enforced on the sliding surfaces, s_1 and s_2 , convergence of the observer is attained and the error dynamics becomes asymptotically stable. In order to examine the stability of the error dynamics, a positive definite Lyapunov function was defined and the time derivative of this function along solutions of the system was shown to be negative definite [14].

6. Theory of sliding mode control

The sliding mode control design procedure consists of two steps. The *first step* involves the selection of a discon-

tinuity surface (sometimes referred to the switching surface or equilibrium manifold) $\sigma(\lambda) = 0$, in the state space, where the control undergoes discontinuities. This surface is determined so that the system behaves in a desired manner (i.e. has the desired eigenvalues).

In *step two*, the continuous control functions $u_d^+(\gamma, t)$ (for $\sigma > 0$) and $u_d^-(\gamma, t)$ (for $\sigma < 0$) are selected such that the control law satisfies the existence conditions for sliding mode.

Consider the nonlinear system in (17)

$$\ddot{\gamma} = F(\gamma) + B(\gamma)u + h(\gamma, t) \quad (17)$$

$$|F(\gamma)| \leq f \quad \text{and} \quad |B(\gamma)| \leq b, \quad |h(\gamma, t)| \leq h_0$$

$F(\gamma)$ is a possibly nonlinear function whose parameters are not exactly known, and its upper bound f is known and is positive. The control is u , $h(\gamma, t)$ is an unknown disturbance with an upper bound h_0 , and B is the control vector, which is not known, but its sign is known and it is bounded by its upper bound, b . The output γ is continuously differentiable and is required to track a time vary reference trajectory, $\gamma_d(t)$, whose higher derivatives exist.

Let the tracking error be

$$\tilde{\gamma} = \gamma - \gamma_d \quad (18)$$

and the initial condition of γ_d be such that

$$\gamma_d(0) = \gamma(0)$$

Hence, tracking could be achieved using a finite control. In order to achieve asymptotic tracking such that $\gamma(t) \equiv \gamma_d(t)$, an equilibrium manifold $\sigma = 0$, is defined in R^n as [8]

$$\sigma(\tilde{\gamma}, t) = \left(\frac{d}{dt} + \beta \right)^{n-1} \tilde{\gamma} \quad \text{and} \quad \beta > 0 \quad (19)$$

$$\sigma = \dot{\tilde{\gamma}} + \beta \tilde{\gamma} \quad (20)$$

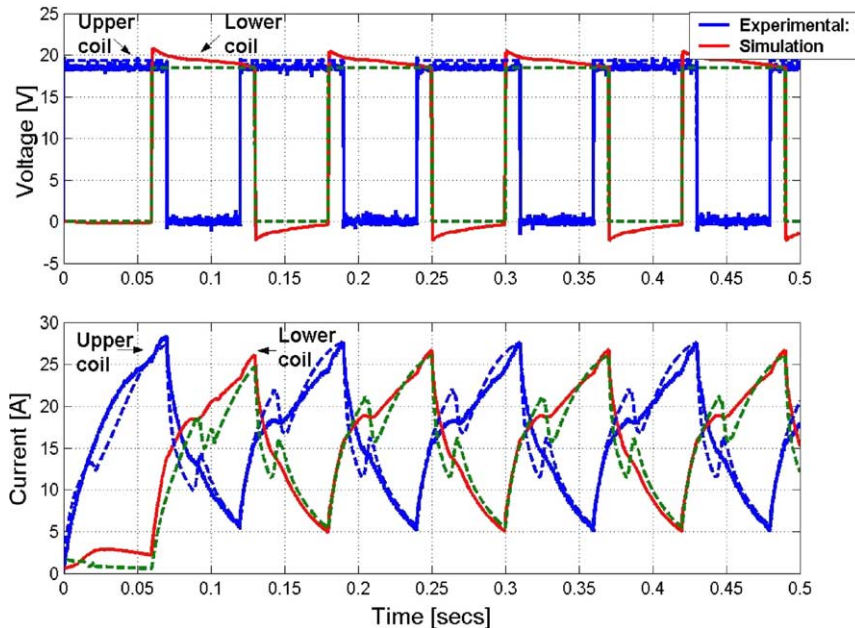


Fig. 11a. Experimental validation of voltage and current.

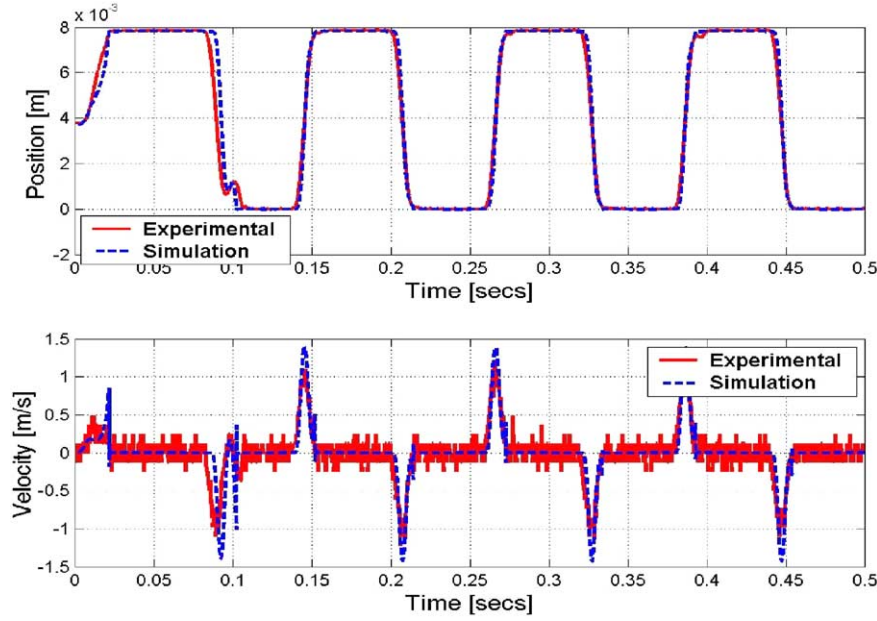


Fig. 11b. Experimental validation of position and velocity.

In the sliding mode; the motion equation coincides with that of the manifold

$$\sigma(\tilde{\gamma}, t) = 0 \Rightarrow \dot{\tilde{\gamma}} = -\beta\tilde{\gamma} \quad (21)$$

The motion equation (21) is of reduced order (in this case a first order differential equation), with solution,

$$\tilde{\gamma}(t) = \tilde{\gamma}(t_1)e^{-\beta(t-t_1)} \quad (22)$$

Motion in sliding mode is asymptotically stable and the error trajectories will reach the equilibrium manifold at time t_1 , and will be confined to this manifold for $t > t_1$. Hence, $\sigma(\tilde{\gamma}, t) = 0$ becomes an invariant set.

The positive scalar, β is chosen to determine the desired rate of convergence of trajectories on to the manifold. The motion in sliding mode is independent of the lumped disturbance term $h(\gamma, t)$ if the *matching conditions* are met. This means that $h(\gamma, t)$ should belong to the range space of $B(\gamma)$. Thus, control, u is designed as a discontinuous control term

$$u = -K \text{sign}(\sigma) \quad \text{for } K > 0 \quad (23)$$

Multiplying (19) by its derivative yields

$$\sigma\dot{\sigma} = (f + h_0 + \beta\tilde{\gamma})\sigma - bK|\sigma|$$

Selecting

$$K > f + h_0 + \beta|\tilde{\gamma}| \quad (24)$$

Hence, the existence and matching conditions are satisfied and sliding mode is enforced despite parametric uncertainties. Thus, robustness is guaranteed on the equilibrium manifold.

In this paper, the control objective is to control the nonlinear system by designing and observer based sensorless, nonlinear controller. This means that only the current will

be measured, and information of the position and velocity will be estimated. The estimate of the position will be used for seating velocity control by feeding back the estimate of the position. Seating velocity control simply means the plunger velocity is reduced to a minimum just before it contacts the face of the electromagnet's core. This is necessary for noise reduction, bounce and fatigue. Since feedback control is not necessary throughout the plunger motion during each cycle, except just before seating, the control strategy will be defined in terms of a two degree of freedom switching controller. This means each actuator will have a controller assigned to it. Initially, an open loop voltage will be the control signal that will drive the plunger from the beginning of travel to a predetermined point just before seating. At this instant, the feedback controller assigned to regulate the seating velocity will receive an external command to switch on and it will output a bounded control intended to force the estimate of the position to track a desired reference trajectory with a low seating velocity. This then involves the designing of a smooth trajectory and the control problem is then reduced to asymptotic regulation of tracking error dynamics in finite time.

7. Sensorless controller design for EMV

The dynamics to be controlled is

$$\begin{aligned} \dot{\hat{z}}_1 &= \hat{z}_2 \\ \dot{\hat{z}}_2 &= \frac{1}{M_t} (-2K_s\hat{z}_1 - B\hat{z}_2 - C \text{sgn}(\hat{z}_2) - \hat{F}_h - \hat{F}_{md} + \hat{F}_{mu}) \\ \dot{\hat{\lambda}}_u &= -f_{3u}(\hat{\lambda}_u, \hat{x}_u, V_{Lu})R + V_{su} \\ \dot{\hat{\lambda}}_d &= -f_{3d}(\hat{\lambda}_d, \hat{x}_d, V_{Ld})R + V_{sd} \end{aligned} \quad (25)$$

Defining the tracking errors for the upper electromagnet as

$$e_1 = r_u - \hat{z}_1, \quad \dot{e}_1 = e_2, \quad \ddot{e}_1 = \ddot{e}_2 \quad (26)$$

and for the lower electromagnet as

$$\bar{e}_1 = -(r_d - \hat{z}_1), \quad \dot{\bar{e}}_1 = \bar{e}_2, \quad \ddot{\bar{e}}_1 = \ddot{\bar{e}}_2 \quad (27)$$

Without loss of generality, only the tracking error dynamics of the upper electromagnet will be analyzed due to the symmetrical nature of the EMV. The reference trajectory was chosen as to have a low seating velocity. The function chosen for the upper coil is

$$r_u(t) = -x_{\max} \exp\left(-\frac{\rho t^2}{2}\right) + x_{\max} + r_u(0) \quad (28)$$

The function for the reference trajectory of the lower coil was chosen as

$$r_d(t) = x_{\max} \exp\left(-\frac{\rho t^2}{2}\right) - x_{\max} - r_d(0) \quad (29)$$

The parameter $r(0)$ is the initial condition and the constant, ρ is chosen to give the desired seating velocity. Large values of ρ increase the rate of convergence of the trajectory, hence the seating velocity. Subtracting (10) from (25) yields the tracking error dynamics as

$$\begin{aligned} \dot{e}_1 &= e_2 \\ \dot{e}_2 &= \frac{1}{M_t} [-2K_s e_1 - B e_2 - C \operatorname{sgn}(\dot{r}_u - e_2) \\ &\quad - F_{mu} + F_{md} + 2K_s r_u + B \dot{r}_u] + \ddot{r}_u \\ \dot{e}_3 &= \Gamma - B_d(\hat{z}_1) R f_{3d} - B_u(\hat{z}_1) R f_{3u} + B_d(\hat{z}_1) V s_d + B_u(\hat{z}_1) V s_u \end{aligned} \quad (30)$$

where

$$\frac{d}{dt}(C \operatorname{sgn}(\dot{r}_u - e_2)) \cong \frac{d}{dt}(C) = 0 \quad (31)$$

$$\Gamma = \left| \frac{1}{M_t} [-2K_s e_2 - B e_3 + 2K_s \dot{r}_u + B \ddot{r}_u] + \ddot{r}_u \right| \quad (32)$$

$$B_u(\hat{z}_1) = 2m_2 \hat{\lambda}_u + m_1 \quad \text{and} \quad B_d(\hat{z}_1) = 2m_2 \hat{\lambda}_d + m_1 \quad (33)$$

The parameters m_1 and m_2 are polynomial coefficients of (8) and (9). The control vectors $B_u(\hat{z}_1)$ and $B_d(\hat{z}_1)$ are positive and nonsingular ($m_i > 0$, $\lambda > 0$) and are dependent on the air-gap estimate, hence on the position estimate. The equilibrium manifolds for the upper and lower electromagnets, respectively are defined as

$$\sigma_1(e_1) = c_1 e_1 + c_2 e_2 + c_3 e_3 \quad (34)$$

$$\sigma_2(\bar{e}_1) = c_4 \bar{e}_1 + c_5 \bar{e}_2 + c_6 \bar{e}_3 \quad (35)$$

The parameters, $c_1 - c_6$ are designed to achieve the desired rate of convergence of state trajectories on to the sliding surfaces. The objective is to design a control that would force the error trajectories to be oriented towards the equilibrium manifold. The conditions for the trajectories to converge on the manifolds, $\sigma_1 = 0$ and $\sigma_2 = 0$ and for sliding mode of exist in these manifolds may be derived based

on a Lyapunov function candidate. Defining a control Lyapunov function for the upper coil as

$$V_u(\sigma_1) = \frac{1}{2} \sigma_1^T \sigma_1 \quad (36)$$

and for the lower coil as

$$V_d(\sigma_2) = \frac{1}{2} \sigma_2^T \sigma_2 \quad (37)$$

Without loss of generality, analysis will be done only for the upper coil, due to the symmetry of the EMV. When the upper coil controller is turned on, $V_{sd} = 0$, and vice versa, but f_{3d} may not be zero due to the large electrical time constant. Differentiating (36) along trajectories of the system

$$\dot{V}_u(\sigma_1) = \sigma_1^T \dot{\sigma}_1 = \sigma^T (c_1 \dot{e}_1 + c_2 \dot{e}_2 + \dot{e}_3) \quad (38)$$

Since feedback control is switched on when $x \neq 0$ and $f_{3d} \neq 0$, it is assumed that $B_d(\hat{z}_1)$ and $B_u(\hat{z}_1)$ are nonsingular, hence control can be chosen as a discontinuous function of the error dynamics. Hence, the control is designed as

$$V_{su} = -U_u \operatorname{sign}(\sigma_1) \quad (39)$$

where, U_u is a scalar gain. The derivative of the Lyapunov function becomes

$$\begin{aligned} \dot{V}_u(\sigma_1) &= \sigma_1^T [c_1 \dot{e}_1 + c_2 \dot{e}_2 + \Gamma - B_d(\hat{z}_1) R f_{3d} \\ &\quad - B_u(\hat{z}_1) R f_{3u} - B_u(\hat{z}_1) U_u \operatorname{sign}(\sigma_1)] \\ &\leq \|\sigma_1^T\| [\|c_1 \dot{e}_1 + c_2 \dot{e}_2 + \Gamma - B_d(\hat{z}_1) R f_{3d} \\ &\quad - B_u(\hat{z}_1) R f_{3u}\| - B_u(\hat{z}_1) U_u \operatorname{sign}(\sigma_1)] \end{aligned} \quad (40)$$

If U_u is selected such that

$$U_u > B_u(\hat{z}_1)^{-1} \|c_1 \dot{e}_1 + c_2 \dot{e}_2 + \Gamma - B_d(\hat{z}_1) R f_{3d} - B_u(\hat{z}_1) R f_{3u}\| \quad (41)$$

Then

$$\sigma_1 \dot{\sigma}_1 < 0 \quad \text{and} \quad \dot{V}_u(\sigma_1) < 0 \quad (42)$$

Similarly, if the control law of the lower coil is chosen as

$$V_{sd} = -U_d \operatorname{sign}(\sigma_2) \quad (43)$$

and (37) is differentiated along trajectories of the system, then U_d can be selected as

$$U_d > B_d(\hat{z}_1)^{-1} \|c_1 \dot{e}_1 + c_2 \dot{e}_2 + \Gamma - B_d(\hat{z}_1) R f_{3d} - B_u(\hat{z}_1) R f_{3u}\| \quad (44)$$

Then

$$\sigma_2 \dot{\sigma}_2 < 0 \quad \text{and} \quad \dot{V}_d(\sigma_2) < 0 \quad (45)$$

If (41)–(45) hold, the trajectories will reach the surfaces in finite time and sliding mode will be enforced in the system. Hence, motion on the surfaces, $\sigma_1 = 0$ and $\sigma_2 = 0$ will be asymptotically stable. For practical stabilization of the error dynamics, the *sign* function, was replaced by sat, the saturation function (46). This is to avoid undesired

finite frequency oscillations of the state vector caused by system imperfections [8]. This is known as *chattering*. The saturation function is defined as

$$\text{sat}\left(\frac{\sigma}{\phi}\right) = \begin{cases} +1 & \frac{\sigma}{\phi} \geq +1 \\ \frac{\sigma}{\phi} & 0 < \frac{\sigma}{\phi} < +1 \\ 0 & \frac{\sigma}{\phi} \leq 0 \end{cases} \quad (46)$$

where ϕ is a boundary layer around the manifold. System trajectories are confined to a ϕ -vicinity of the sliding man-

ifold $\|\sigma_1\| \leq \phi$ and $\|\sigma_2\| \leq \phi$ instead of exactly on $\sigma_1 = 0$ and $\sigma_2 = 0$, respectively, as suggested in ideal sliding mode. Hence, the trajectories in the vicinity of σ_1 and σ_2 will be uniformly ultimately bounded [9].

8. Simulation results

The observer in (15) was implemented in simulation with the validated model of the EMA, using numerical integration in Simulink. The observers converged asymptotically and the dynamics on the sliding surfaces were asymptotically stable (Figs. 12 and 13).

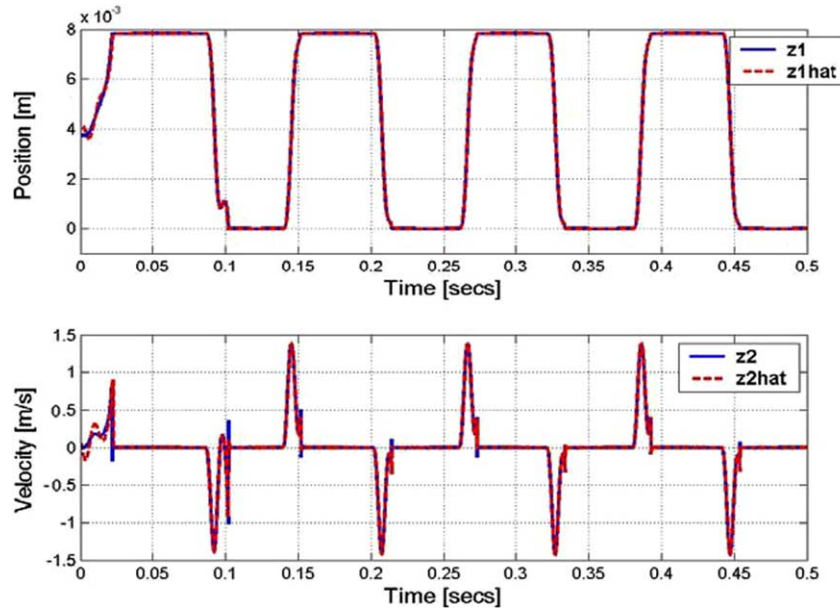


Fig. 12. Open loop observer simulation: position and velocity.

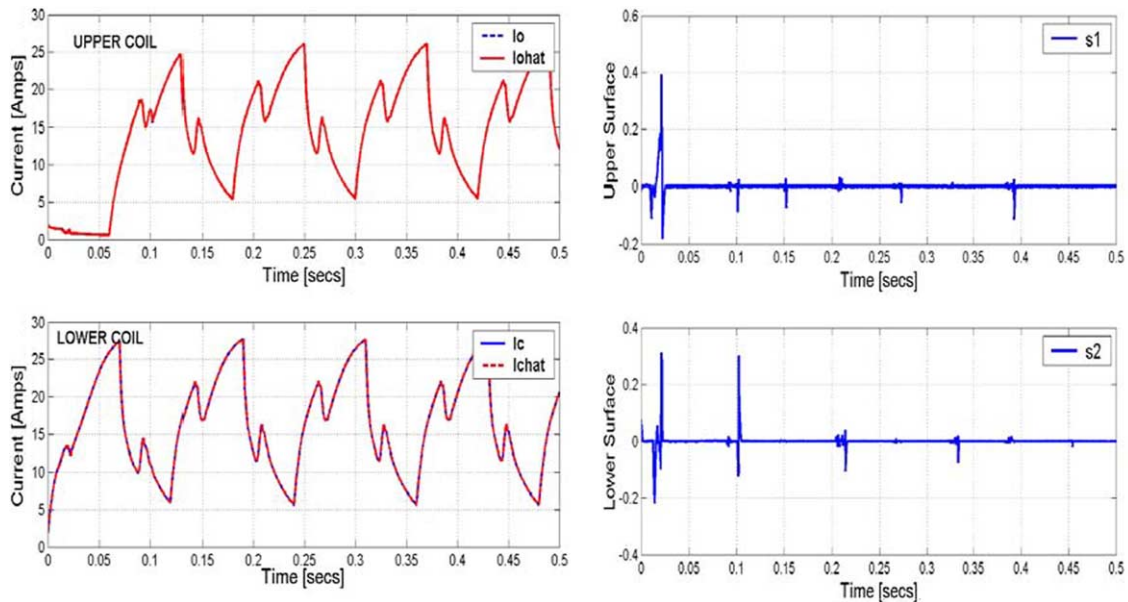


Fig. 13. Open loop observer simulation: current (left) and sliding surfaces (right).

The control designed for (25) was implemented in a simulation with the validated model of the EMV. The control strategy was implemented as a combination of closed and open loop control. The controller is embedded within a triggered subsystem. The open loop control usually is used to accelerate the plunger's inertia and just before the plunger contacts any of the electromagnets, closed loop control is switched-on for as long as the actuator is required to attract and hold the plunger. When the command to release the plunger is received, closed loop control is switched off and open loop control is turned on to accelerate the plunger towards the other electromagnet and the cycle

is repeated. Fig. 14 shows that the position estimate converged to its actual value and the desired trajectory was closely tracked by the position and its estimate.

Comparing the controlled and the uncontrolled case, feedback control reduced the seating velocity from about 0.5 m/s in the uncontrolled case, to about 0.065 m/s in the controlled case (Fig. 15). Less current was consumed when feedback is used than in open loop (Fig. 16). The average amount of current used for seating is about 11.0 A.

Feedback was switched on at 0.09 s and at 0.15 s (Fig. 17) for the upper and lower electromagnets, respec-

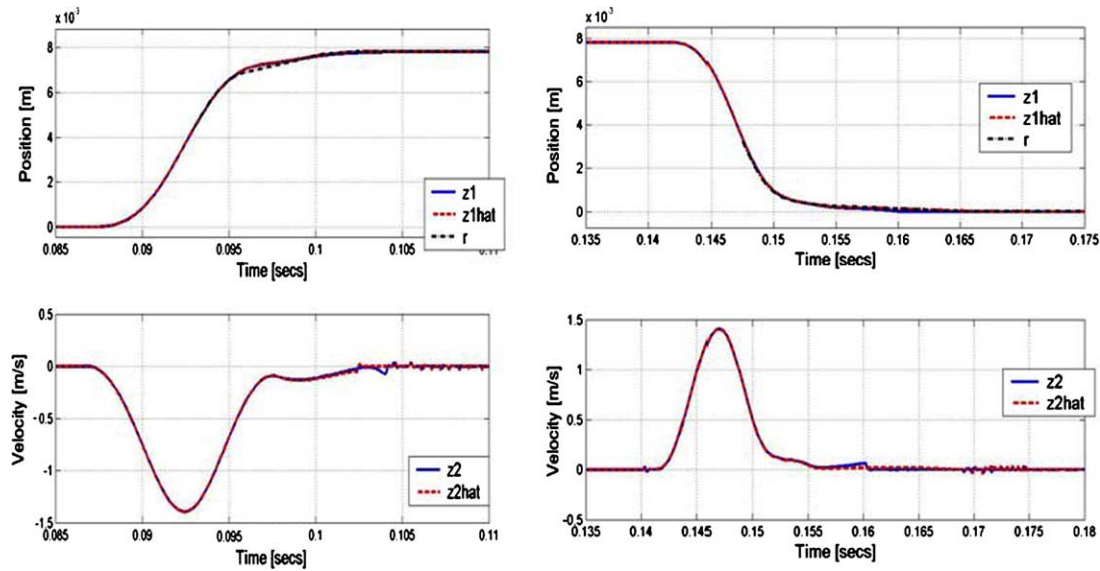


Fig. 14. Position tracking control: upper coil (right) lower coil (left).

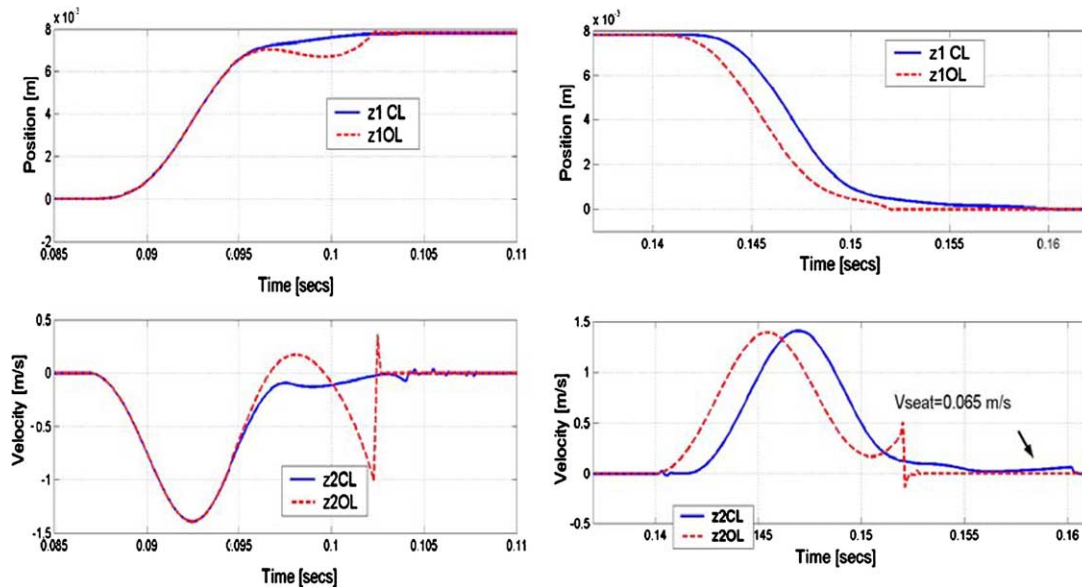


Fig. 15. Controlled (—) and uncontrolled (---): upper Coil (right), lower coil (left).

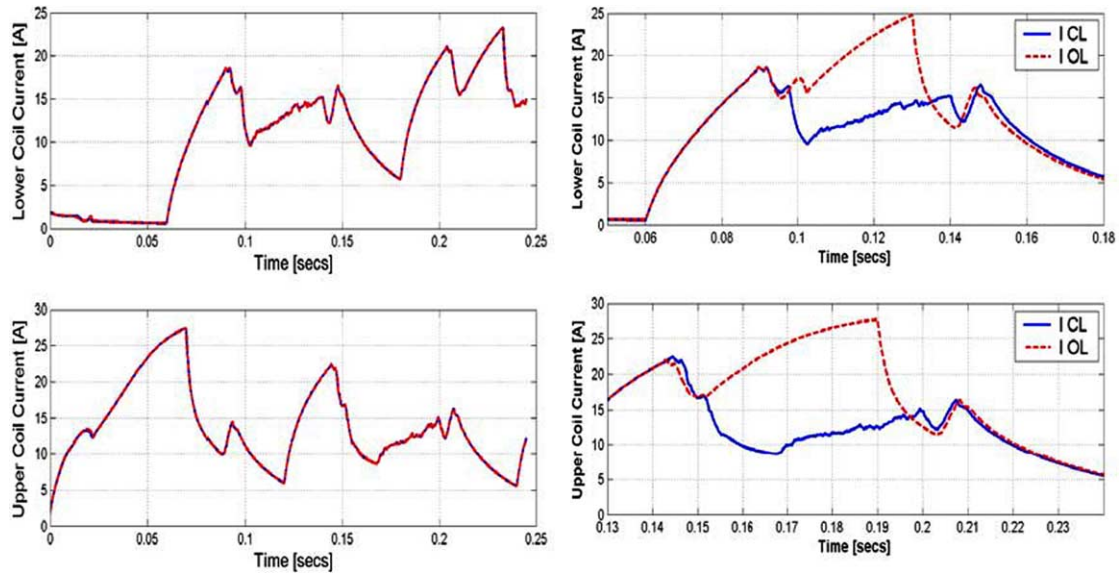


Fig. 16. Current and its estimate in closed loop (left). Comparing closed loop (—) and open loop (---) current (right).

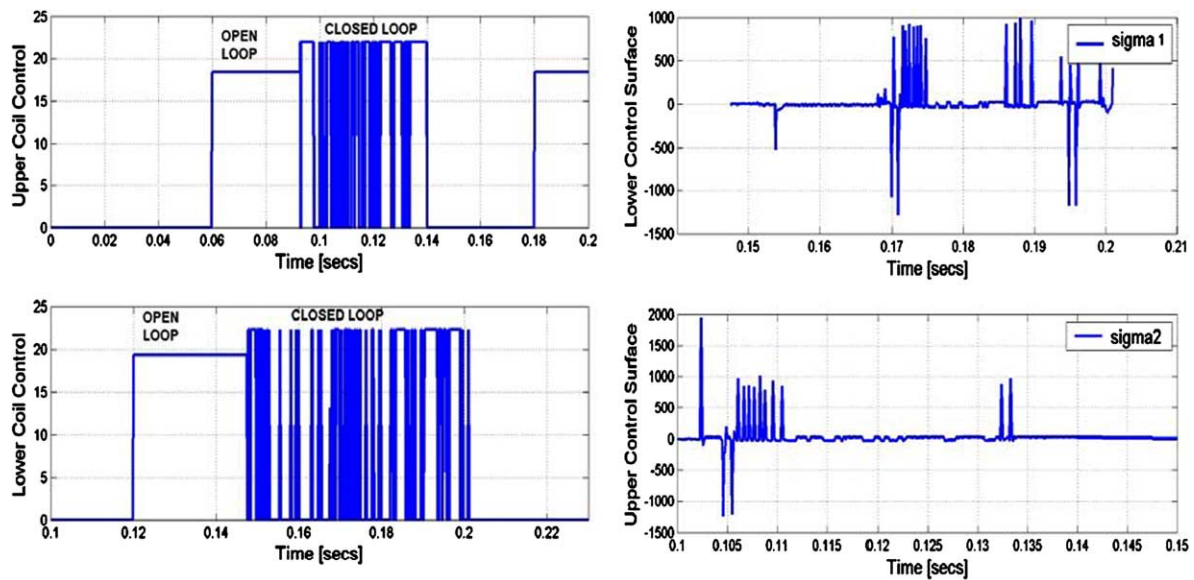


Fig. 17. Closed loop control voltage (Volts) (left). Sliding surface (right).

tively. After these periods, dynamics on the surfaces converged to the origin and sliding mode was enforced on both surfaces.

9. Experimental results

9.1. Equipment

Open loop and closed loop control implementation was done in the Dspace 1103 Digital Signal Processor (DSP) rapid prototyping system at a sampling rate of 0.00015 s. This sampling rate was chosen due to hardware constraints. The Dspace board then generated command volt-

ages into a servo-amplifier. The position of the plunger was measured by a high resolution Microtrak 700 Laser sensor just to validate the accuracy of the position observer. The current was measured by a closed loop current sensor. The experimental setup is shown in Fig. 18.

9.2. Open loop control (OL)

The current, position and velocity observers were validated experimentally in open loop. The estimates matched their actual values closely, and the bounce of the plunger was captured (Fig. 18). Actuator bounce is responsible for the peaks in the velocity estimate. Sliding mode was

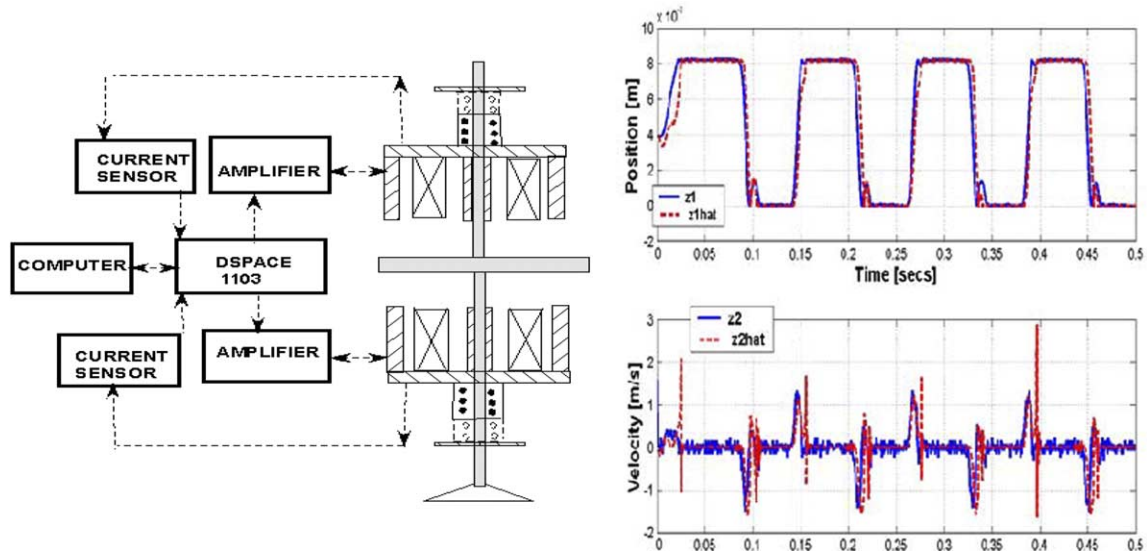


Fig. 18. Experimental setup (left), position and velocity observer validation (right).

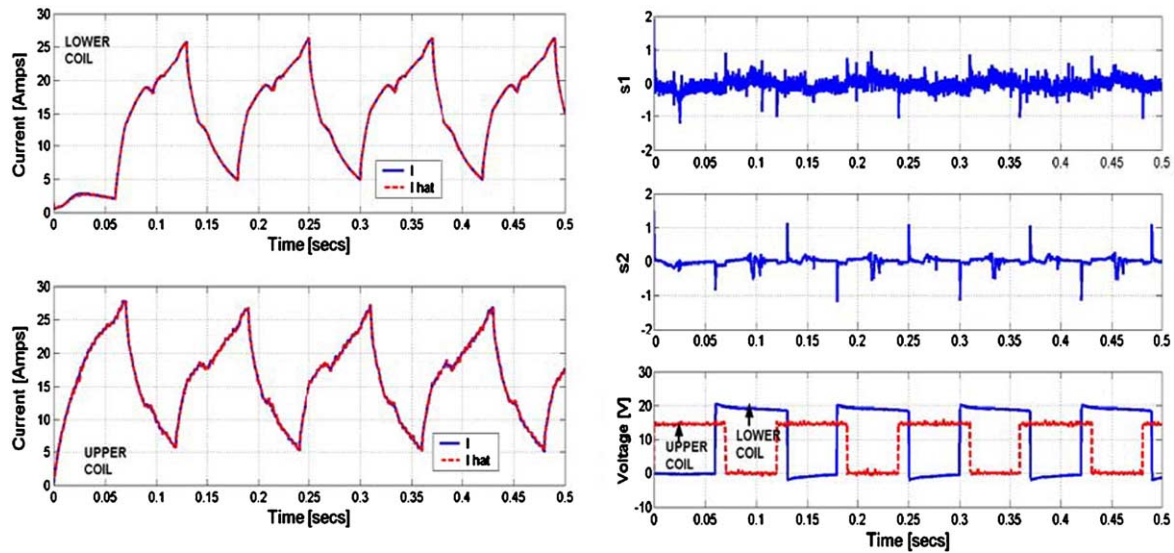


Fig. 19. Current observer (left), sliding surfaces and open loop voltage (right).

enforced on both surfaces (Fig. 19). The input voltage was 15 V and 20 V respectively, for the upper and lower coils.

9.3. Closed loop control (CL)

The metal-to-metal contact initiated when either electromagnet catches the plunger causes a loud “banging” noise. Reducing the noise means slowing down the plunger before it contacts either electromagnet, i.e. at opening and at seating. This was achieved by switching on each controller just before contact is made between the plunger and the face of either electromagnet to force the position estimate to track a desired trajectory with low seating velocity (Fig. 20).

Tracking was achieved but becomes increasingly difficult with reduction in air-gap. Bounce is eliminated when contact is made with the lower electromagnet and the seating velocity is reduced from 0.8 m/s (uncontrolled) to 0.3 m/s (Fig. 21). Less current was used in closed loop control than in open loop (Fig. 22). The best seating velocity obtained was 0.05 m/s (Fig. 23) and this was possible due to good tracking control, but estimation error just before seating was comparatively greater than the case shown in Fig. 20.

Closed loop repeatability tests were performed to observe the consistency of the controller and observer performances for different runs (Fig. 24). This was implemented by controlling both electromagnets. Seven separate

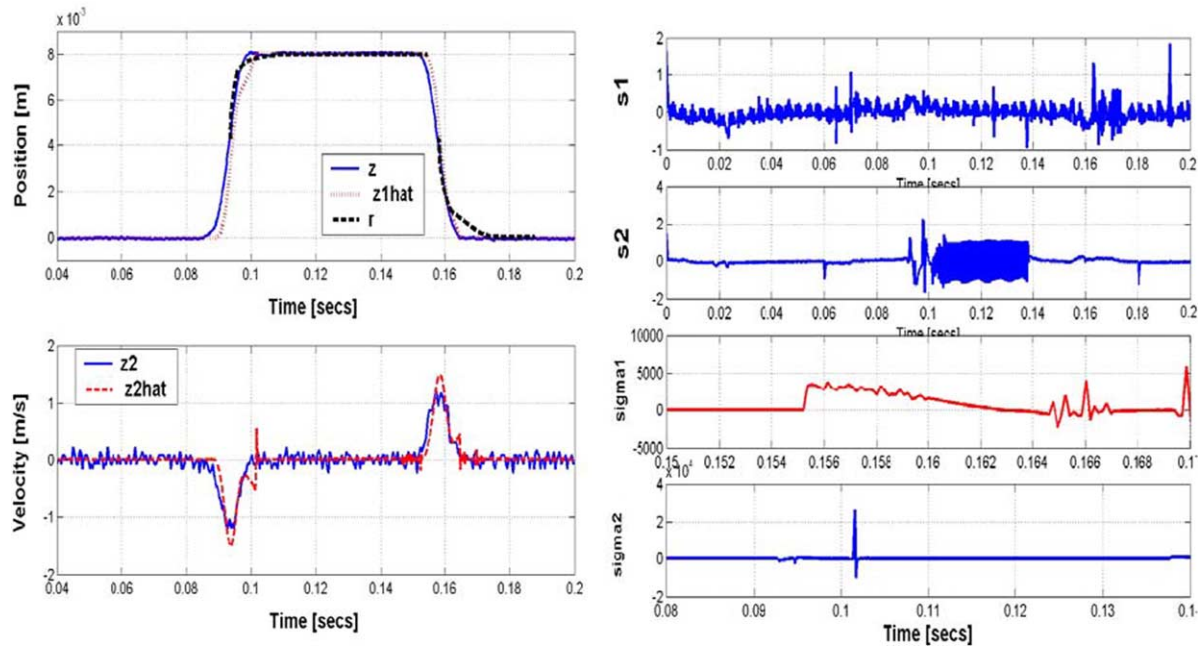


Fig. 20. Tracking control (left) and sliding surface (right).

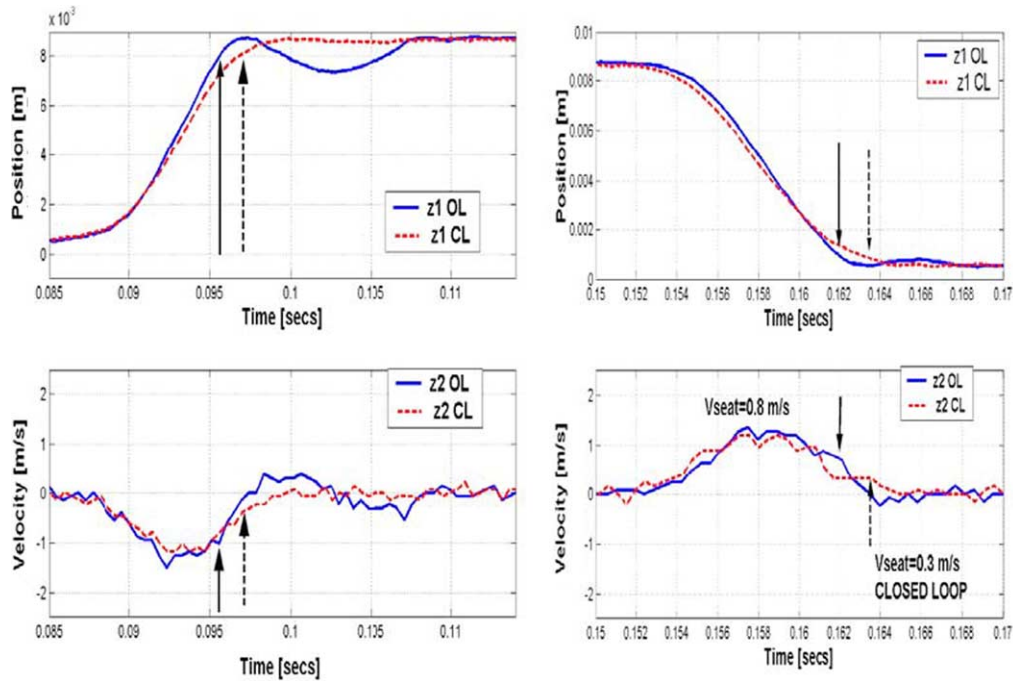


Fig. 21. Open loop (OL) vs. closed loop control (CL): lower coil (left), upper coil (right). Note: Arrows indicate instant just before seating.

runs were conducted and the seating velocity is reduced in all cases and values ranged from 0.35 m/s to 0.05 m/s. Having a low seating velocity neither depends on asymptotic tracking of the desired trajectory nor on a small estimation error.

However, certain factors were responsible for control inaccuracies, namely:

- The spring pre-compression displacement changes after several high frequency cycles. This is due to the fact that the spring rate is not constant, but a complex nonlinear function of several variables.
- Also control inaccuracies just before seating was caused by the control discontinuity introduced into the observer when switching from open loop to closed loop control.

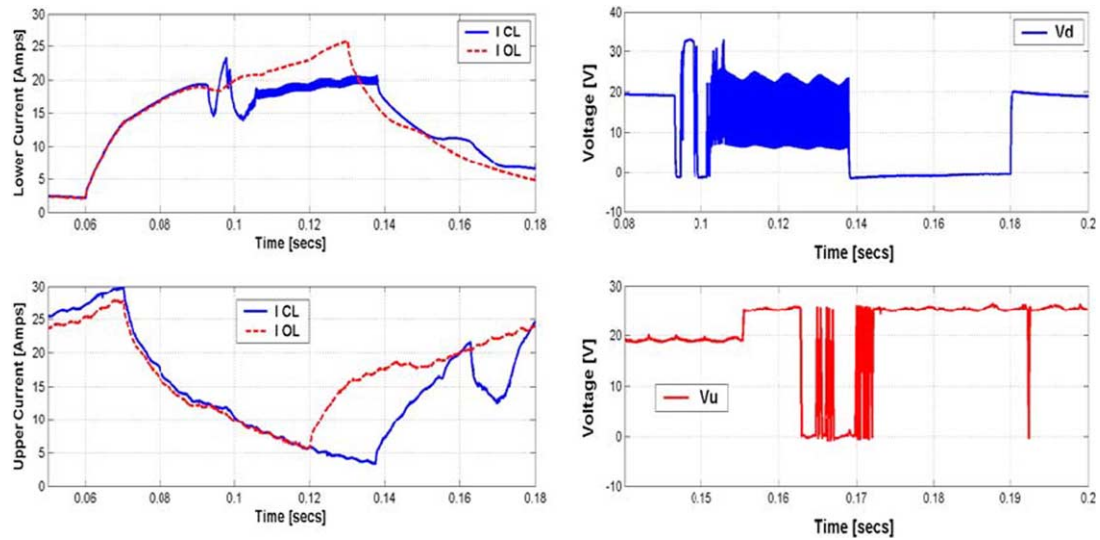


Fig. 22. Current: open loop (OL) vs. closed loop control (CL) (left). Control voltage: closed loop (right).

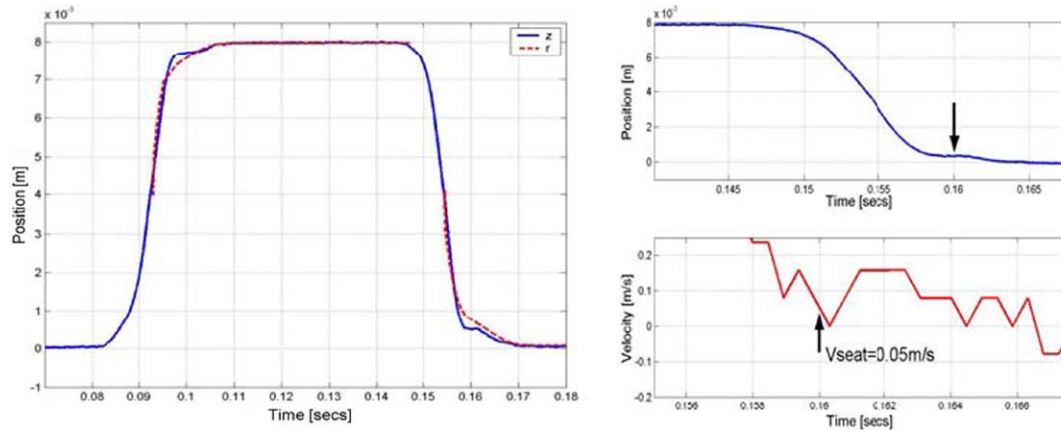


Fig. 23. Desired trajectory and actual position (left), position (right-top), seating velocity (right-bottom).

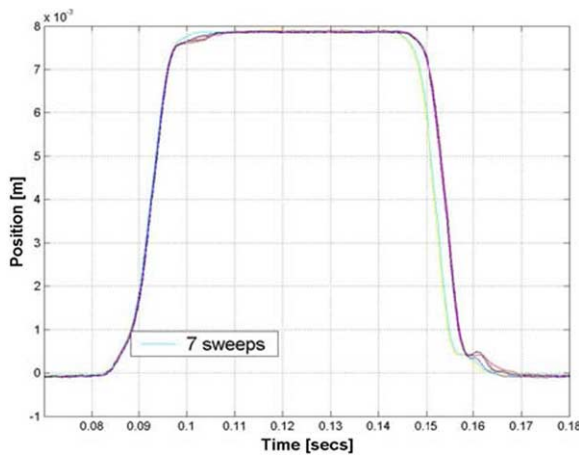


Fig. 24. Repeatability test on actual position in closed loop (7 sweeps).

processor. This adds a significant computational load on the processor and this acts as a hardware constraint in sampling rate selection. Also, this introduces lag into the feedback loop, reducing phase margin. Hence, the choice of sampling rate was limited by hardware constraints.

10. Conclusions

A nonlinear model for the EMV was presented and validated experimentally. Transient characteristics of the EMV were taken into account in the model, including saturation, hysteresis, bounce and mutual inductance. A sliding mode based “sensorless” control strategy was implemented in simulation and experimentally to reduce impact noise and seating velocity, with a peak voltage of 30 V and an average voltage of 25 V. The minimum seating velocity was 0.05 m/s for an event time of 22 ms. The state estimates converged to their actual values asymptotically.

- Implementing an observer based sensorless controller implies the entire model of the EMV is running on the

Repeatability was satisfactory and the actual position followed the desired trajectory quite well.

Acknowledgements

The research conducted in this study was supported by the *General Motors Research Center*. In particular, the authors greatly acknowledge the technical support provided by *Zongxuan Sun* and *Patrick Usoro* of the *GM Research Center*.

References

- [1] Podanar D et al. Characteristics of a small engine equipped with and electromagnetic valve actuation system. SAE paper 981908, 1998.
- [2] Tai C, Tsao T-C. Quiet seating control design of an electromagnetic engine valve actuator. ASME 2001.
- [3] Wang Y, Stefanopoulou A. Modeling and control of electromechanical valve actuator. SAE paper 2002-01-1106, 2002.
- [4] Butzmann J, Melbert J, Koch A. Sensorless control of electromagnetic actuators for variable valve train. SAE paper 2000-01-1225, 2000.
- [5] Domanski, S. Effects contributing to the temperature rise of electromagnetic actuators. SAE paper 2000-01-0543, 2000.
- [6] Hofmann W, Stefanopoulou A. Valve position tracking for soft landing of electromechanical camless valvetrain. In: Proceedings on IF AC workshop on automotive control, Karlsruhe, March 2001, p. 305–310.
- [7] Slotine JJ, Hedrick JK, Misawa EA. On sliding observers for nonlinear systems. *J Dyn Syst Measur Control* 1987;109:245–52.
- [8] Slotine J-J, Li W. *Applied nonlinear control*. Prentice-Hall; 1991.
- [9] Utkin V, Guldner S. *Sliding mode control in electromechanical systems*. Taylor and Francis; 1999.
- [10] Lequesne B. Fast-acting, long-stroke solenoids with two springs. *IEEE Trans Indus Appl* 1990;26(5).
- [11] Longstaff K et al. Internal combustion engines. US Patent No. 3,882,833, May 13, 1975.
- [12] Buchal J. Method of activating an electromagnetic positioning means and apparatus for carrying out the method. US Patent No. 5,544,986, October 1, 1985.
- [13] Pischinger F, Kreuter P. Arrangements for electromagnetically operated actuators. US Patent No. 4,515,343, May 7, 1985.
- [14] Eyabi P. Modeling and sensorless control of solenoidal actuators. Ph.D. dissertation, The Ohio State University, June 2003.
- [15] Vaughan ND, Gamble JB. The modeling and simulation of a proportional solenoid valve. *Trans ASME* 1996;118.
- [16] Inman DJ. *Engineering vibration*. 2nd ed. Prentice-Hall; 2000.